# SO cross-cl pipeline: decision tree v3 — Bayesian search + profile scheme

*left: search for moments with the evidence · right: frequentist branch, profile scheme*

## A · Bayesian search for moments (evidence, r free)

run BBPower without moments → $M_{0,r}$

is $\chi^2$ good? (PTE > 0.01)
- No → Res_wr: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families of moments, each with its own β free
- Yes → is r compatible with zero at 1σ?
  - No → Res_wr: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families of moments, each with its own β free
  - Yes → **upper limit on r ($M_{0,r}$)**

moment detected? (**) 0 outside the 68% HPD of an amplitude
- Yes → $M_{1,r}$ = $M_{0,r}$ + a detected family, in order of residual evidence (***); full fit (all parameters free), pruned (****)
- No → Res_r0: residuals of $M_{0,0}$ (r = 0) all parameters fixed, own β free; test the families of moments d, s, dT, d2, s2, ds1, ds2

is $M_{1,r}$ preferred over $M_{0,r}$? (ln B > 0)
- Yes → $M_{0,r}$ = $M_{1,r}$, start again
- No → moment detected? (**) 0 outside the 68% HPD of an amplitude
  - Yes → $M_{1,r}$ = $M_{0,r}$ + the detected family with the highest residual evidence (***); full fit (all parameters free), pruned (****)
  - No → $M_{1,0}$: same family, r = 0, pruned (****)

is $M_{1,r}$ preferred over $M_{0,r}$? (ln B > 0)
- Yes → $M_{0,r}$ = $M_{1,r}$, start again
- No → is $M_{1,0}$ preferred over $M_{0,r}$? (ln B > 0)
  - Yes → $M_{0,r}$ = $M_{1,r}$, start again
  - No → $M_{0,r}$ = $M_{1,r}$, start again

(**) all the amplitudes of the family are fitted (emcee); detected if 0 is outside the 68% HPD (highest posterior density interval) of at least one of them; only the amplitudes with 0 outside their HPD are kept
(***) PolyChord evidence of the residual fit (only the family's moments and its β free): only ranks the families
(****) amplitudes with 0 in their 68% HPD (posterior of the full fit) removed, model refitted, until all are detected; kept if ln B(pruned vs full) > -2.5; the pruned model is the one compared with M_0,r

## B · Profile scheme (new): uses r as a profile, not marginalised

M = current model, C = candidate moment · PolyChord only where an evidence is needed (steps 2, 3) · profiles: max likelihood · parameters: emcee

**0 — starting point**

$M_{1,0}$ preferred → M = $M_{1,r}$, the best model found: no candidate, go straight to step 3

a family was tried as $M_{1,r}$ and lost ($M_{1,0}$ not preferred) → M = $M_{0,r}$, candidate C = that family

nothing new detected in Res_r0 → M = $M_{0,r}$, no candidate: go straight to step 4 (r from its profile)

**1 — profile of r of M + C**

r fixed on a grid, all other parameters re-fitted (max likelihood) → r* = minimum

**2 — is M + C preferred over M?**

each model at its own best fit (M + C at r*, M at the minimum of its profile); free only the moment spectra + the same β in both models; ln B > 0
- Yes → M = M + C (C enters the model)
- No → C rejected (not proposed again)

**3 — Res_wr*: residuals of M re-fitted at r = r***

all groups not yet in the model and not rejected (also dT, d2, s2, ds1, ds2): screening with emcee (0 outside HPD 68%), then evidence against the null (same point, moment amplitudes = 0, only β free): passes if ln B > 0

does a new moment pass?
- Yes: C = the best one, back to 1
- No → **4**

**4 — result**: r from the profile of M used as a likelihood × prior of the tree → median, 68%, σ, r_95 (also with r ≥ 0); other parameters: emcee of M with r = r* fixed

0 within 3σ of r?
- Yes → **upper limit**
- No → **r detection**